



Boehmite-coated *h*-BN modified MgO castable and its high-temperature wettability to slag

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Abstract

Hexagonal boron nitride (*h*-BN) neither wets nor reacts chemically with molten metals and slag, demonstrating an excellent corrosion resistance. In this work, a boehmite-coated *h*-BN was prepared *via* hydrothermal method, which subsequently be incorporated into MgO castable for the anti-corrosion of molten slag. Because of the natural hydrophobicity of starting *h*-BN (WCA of $\sim 150^\circ$), the boehmite-coating endowed them with a hydrophilic behavior ($\sim 46^\circ$), which could ensure the homogeneous dispersion of boehmite-coated *h*-BN in MgO slurry. The *h*-BN-modified MgO castables exhibited a superior non-wettability performance than its unmodified counterpart. A high-temperature contact angle of $\sim 33^\circ$ between the slag and modified MgO castable was detected when 2 wt% of *h*-BN was used, which was higher than that of unmodified one ($\sim 19^\circ$). Correspondingly, the slag penetration depth of former case was reduced by 44.1% over the latter one.



Keywords

Boehmite; *h*-BN; MgO castable; High-temperature wettability

1. Introduction

Slag corrosion is the major cause of refractory damage, in which the complex interactions between high-temperature slag and refractories results in wetting, melting and corrosion [[1], [2], [3], [4], [5], [6], [7], [8]]. It is well-known that the slag intrusion (penetration depth) is mainly dominated by the contact angle, porosity of refractory, slag's adhesion work and surface tension [9]. Given by these, enormous efforts have been devoted on regulation of slag wettability on refractory. For instance, Yoon et al. [10] investigated the wetting behavior (1873 K) of CaO-Al₂O₃-SiO₂-MgO slag on Mg-rich and Al-rich spinels, showing that the former contact angle decreased more rapidly than that of latter one, and the permeability of slag increased with the increase of MgO content in spinel. Shen et al. [11] examined the wetting behavior of CaO-Al₂O₃-SiO₂-MgO-CaF₂ slag on dolomitic refractory. When the extension of soaking time was from 1 h to 3 h, the slag contact angle decreased from 10.5° to 5.0°, and the corrosion depth correspondingly increased from 55 mm to 100 mm. Based on these, developing the refractories with enhanced non-wettability to slag holds a far-reaching implication.

Due to its poor slag wettability, carbon-containing refractory can raise the contact angle of slag, and then prevent the further corrosion [[12], [13], [14], [15]]. For example, Liu et al. [16] investigated the slag wetting behavior on MgO and MgO-C refractories using a seated drop method. It showed that the former contact angle decreased from 115° to 35° in 80 s, and remained essentially constant in the subsequent 60 s, while the contact angle of latter one decreased from 116° to 55° at the same test time. In our recent work [17], a lotus leaf-bioinspired carbon nanotube (CNT)-functionalized MgO castable was fabricated using waste polyethylene as carbon source, which exhibited not only “super-slag-phobic” behavior toward slag, but also excellent super-nonwetting performance against molten salt, liquid copper, and glass. Despite these remarkable results, some crucial issues

including poor oxidation resistance and hydrophobicity of carbon still hinder the practical application, especially in the case of castable refractories.

Hexagonal boron nitride (*h*-BN) possesses an analogous structure to graphite, which layers, connected through covalent bonds between B and N atoms, are bonded to each other by *van der Waals* [[18], [19], [20], [21], [22]]. Owing to its excellent oxidation resistance and non-wettability by slag, *h*-BN is also regarded as an attractive substitute for graphite on preparation of carbon-free refractory [[23], [24], [25], [26], [27], [28]]. To respond to this, a boehmite-coating *h*-BN was prepared by hydrothermal route, aiming to improve the water-wettability in MgO castables. Density functional theory (DFT) was used to investigate the wetting mechanism of H₂O on the coated *h*-BN. Finally, *h*-BN-modified MgO castables were prepared, and the physical property, thermal shock resistance and high-temperature wettability to slag were systematically investigated.

2. Experimental procedure

2.1. Raw materials and sample preparation

h-BN (<25 μm, purity >99wt%) was provided by Shandong Zibo Jingyi Ceramics Technology, China. Aluminum chloride hexahydrate (AlCl₃·6H₂O, purity >98wt%), sodium hydroxide (NaOH, purity >96wt%), polyvinylpyrrolidone (PVP, purity >99wt%) and sodium dodecyl benzene sulfonate (SDBS, purity >99wt%) were supplied by Sinopharm Co. Ltd. Shanghai, China.

The suspensions with *h*-BN/AlCl₃·6H₂O molar ratios of 0.5/1, 1/1, 2/1 and 3/1 were mixed and stirred for 30min. Then, 1 molL⁻¹ of NaOH solution was added to generate the Al(OH)₃ precursor, which was subsequently poured into an autoclave, and heated at 463K for 12h. The resultant colloids were dried at 353K after vacuum filtration to obtain the boehmite-coated *h*-BN powders. The detailed processing conditions are listed in Table S1.

The raw materials of MgO castable (Table S2) was first dry-blended for 3min, and 5.2wt% of water was incorporated and wet-mixed for another 3min. Subsequently, the mixture was molded under vibration to prepare the specimens with sizes of 25×25×140mm³. After 24h curing at ambient temperature, the specimens were further dried at 383K for 24h to obtain the final boehmite-coated *h*-BN modified MgO castable.

2.2. High-temperature wetting experiment

A cylindrical slag ingot ($\Phi 5.5 \times 7$ mm) was placed on the surface of samples, and heated in a vacuum furnace at 1573 K with a heating rate of 10 K/min. Afterward, the furnace temperature was further increased to 1973 K under atmosphere of flowing Ar. The wetting behavior between the slag (Table S3) and sample was characterized by recording the dynamic changes of contact angle (θ), diameter (d) and height (h) of the slag droplets *via* sessile drop method (TA-Z16B01, Tianjin Zhonghuan Furnace Company, China).

2.3. Characterization and testing

Phase composition, micromorphology, surface characteristics, and wettability of the coated *h*-BN were analyzed using XRD, SEM, TEM, and TG-DSC, respectively, and their detailed information are provided in the “Supplementary Information.” Apparent porosity and bulk density of specimens were determined according on the Archimedes principle. Cold modulus of rupture (CMOR) and cold compressive strength (CCS) were evaluated using an universal electronic testing machine (EDC 120, DOLI Company, Germany, China). Thermal shock resistance was assessed by three cycles of air quenching from 1373 K to room temperature. Slag corrosion behavior was examined through a static crucible test, which was performed at 1823 K for 3 h in an alumina crucible containing fine coke powder.

2.4. DFT calculation

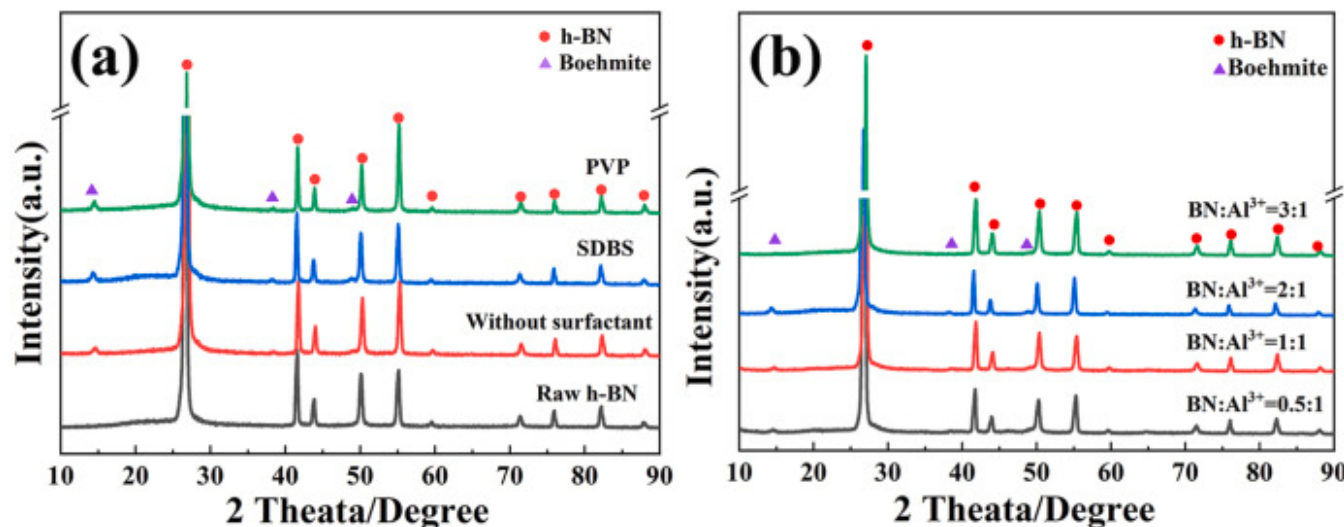
DFT calculations were conducted using the CASTEP module on Materials Studio software. Surface models of *h*-BN (001) were constructed based on the periodic cell of *h*-BN after geometric optimization. A vacuum layers of 10 Å thickness was set to eliminate interactions between neighboring atomic layers. The *h*-BN molecule optimization was carried out within a cubic cell measuring $10 \times 10 \times 10 \text{ Å}^3$. The generalized gradient approximation (GGA) employing the Perdew-Burke-Ernzerhof (PBE) functional was utilized [29,30]. A Monkhorst-Pack *k*-point mesh of $3 \times 3 \times 1$ was applied. Structural relaxation proceeded until the force and energy convergence thresholds reached 0.05 eV/Å and 10^{-4} eV, respectively [31].

3. Results and discussion

3.1. Preparation and characterization of boehmite-coated *h*-BN

Fig. 1a displays the XRD patterns of sample resultant from hydrothermal treatment with the addition of SDBS and PVP. Strong diffraction peaks attributed to *h*-BN (ICDD 026–1076) were detected, accompanied by a broad peak around 15° , indicating the

formation of boehmite (ICDD 074–1895). Use of different surfactants showed no obvious effect on the formation of boehmite. Fig. 1b shows the XRD spectra of samples prepared under different h -BN/ Al^{3+} molar ratios. Distinct diffraction peaks of h -BN and boehmite were observed in all the samples at h -BN/ Al^{3+} molar ratios in the range of 0.5–2.0. As h -BN/ Al^{3+} molar ratio increased to 3, the peak intensity of boehmite became weak due to an excessive addition of h -BN.



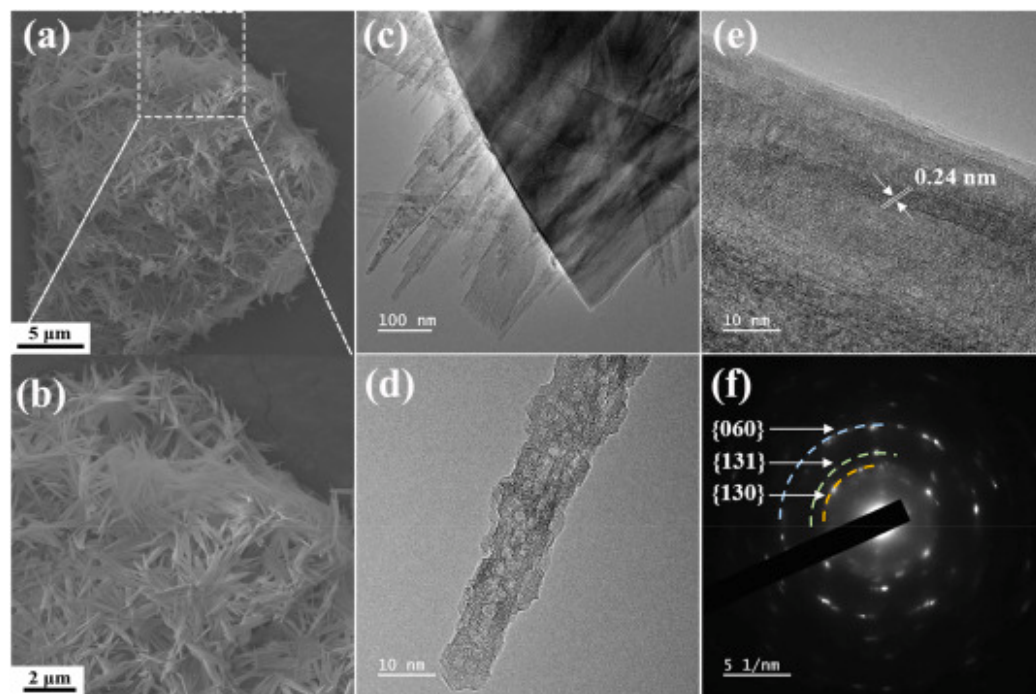
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Fig. 1. XRD patterns of raw h -BN and coated ones prepared (a) with addition of SDBS and PVP, (b) with different h -BN/ Al^{3+} molar ratios.

The morphology of boehmite-coated h -BN was characterized using SEM and TEM, as shown in Fig. 2 and S1–S2. It indicates that the as-formed boehmite presented a fiber-like structure (Fig. 2a–b). Fig. S1 shows the surface of pristine h -BN was smooth, exhibiting a typical lamellar structure (Fig. S1a). Only a few fibers were observed on the h -BN surface after hydrothermal treatment without surfactants, and most of the h -BN sheets were unencapsulated (Fig. S1b). The samples using SDBS surfactant displayed an enhanced surface encapsulation (Fig. S1c). Especially when PVP was used, the long straight fibers were fully encapsulated on h -BN sheet surface (Fig. S1d). Fig. S2 further presents the effect of h -BN/ Al^{3+} molar ratios on the microstructure. At a ratio of 0.5/1 (Fig. S2a), the surface of h -BN was uniformly coated with a layer of boehmite fibers, which were stacked together to form a thicker coating layer. As the h -BN/ Al^{3+} molar ratio was increased to 1/1–2/1 (Fig. S2b–c), the coating layer,

although being significantly thinned, remained uniformly distributed on the *h*-BN surface. When the *h*-BN/Al³⁺ was further increased to 3/1 (Fig. S2d), only a few boehmite fibers were present. The low-magnification SEM images with comprehensive EDX analysis demonstrate that as the *h*-BN content increased, the Al content decreased from 7.32% to 3.41% (Figs. S3–S6). The TEM image in Fig. 2c shows a fibrous material deposited on *h*-BN sheet, further proving an uniform coating of boehmite. The boehmite fibers exhibited a porous structure (Fig. 2d) with a lattice spacing of 0.24 nm (Fig. 2e), corresponding to the (103) crystal plane of boehmite. The selected area electron diffraction (SAED) pattern displayed multiple diffraction rings (Fig. 2f), respectively corresponding to the {131}, {130} and {060} crystal planes of boehmite, which indicated that the boehmite fiber is polycrystalline [32,33].



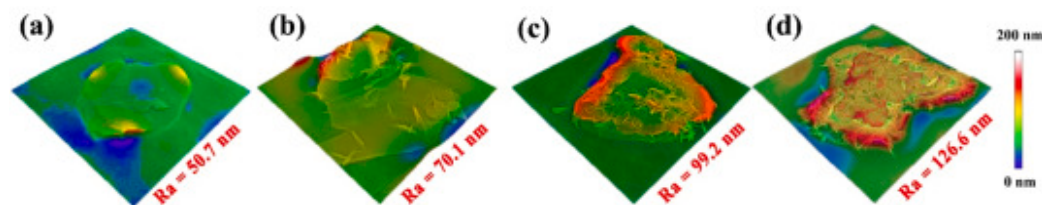
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Fig. 2. (a, b) SEM (c, d) TEM, (e) HRTEM and (f) SAED images of boehmite coated *h*-BN.

Fig. 3 displays the 3D roughness photos of boehmite-coated *h*-BN, whose SEM images are shown in Fig. S1. According to the calculation of average surface roughness, the value of original *h*-BN was only 25 nm (Fig. 2a), while that of the hydrothermally

coated one was 70.1 nm (Fig. 2b). When the surfactants were used, the roughness could be further raised to 99.2 nm (Figs. 2c) and 126.6 nm (Fig. 2d) in the cases of SDBS and PVP, respectively, indicating that the formation of boehmite could improve the surface roughness of *h*-BN. Moreover, the reduction of surface roughness would also significantly decrease the Al content (1.67%, Fig. S7) due to the use of SDBS.

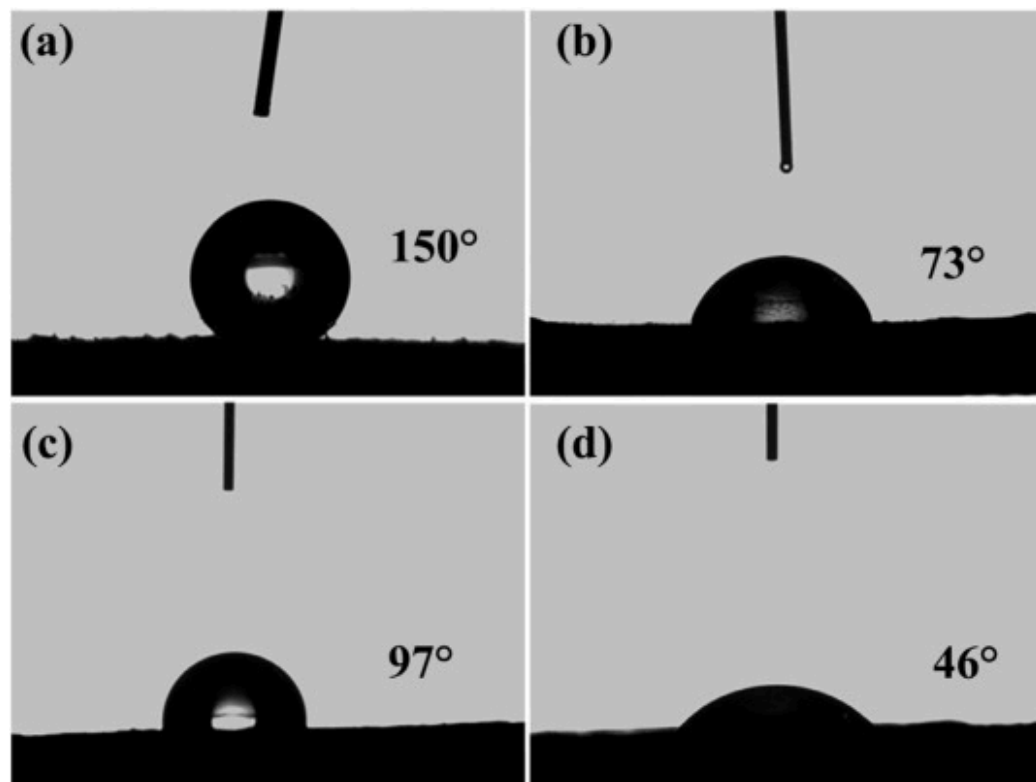


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Fig. 3. Surface roughness of (a) raw *h*-BN, and coated ones prepared (b) without surfactants, and with (c) SDBS and (d) PVP.

Fig. 4 exhibits the WCAs of original *h*-BN and boehmite-coated ones. The WCAs of original sample was 150° (Fig. 4a), while the value of the composite powder without surfactant was decreased to 73° after hydrothermal treatment, suggesting that the boehmite-coating significantly enhanced the wettability of *h*-BN (Fig. 4b). When further using SDBS, the WCA was increased to 97° on the contrary (Fig. 4c). It could be attributed to: 1) the attachment of hydrophilic sulfonate group on *h*-BN or boehmite, in which the hydrophobic long-chain alkyl group was oriented outward, resulting in a weakened water wettability (Fig. S8) [34,35], and 2) the significant reduction of boehmite content (1.67%, Fig. S7) on the sample surface using SDBS. Notably, the WCA value was significantly decreased to 46° (Fig. 4d) in the case of PVP, which could be due to the presence of hydrophilic carbonyl and pyrrole ring groups of PVP. Fig. S9 presents the WCAs of boehmite-coated *h*-BN with different *h*-BN/Al³⁺ molar ratios, demonstrating that the WCAs increased along with the molar ratio.



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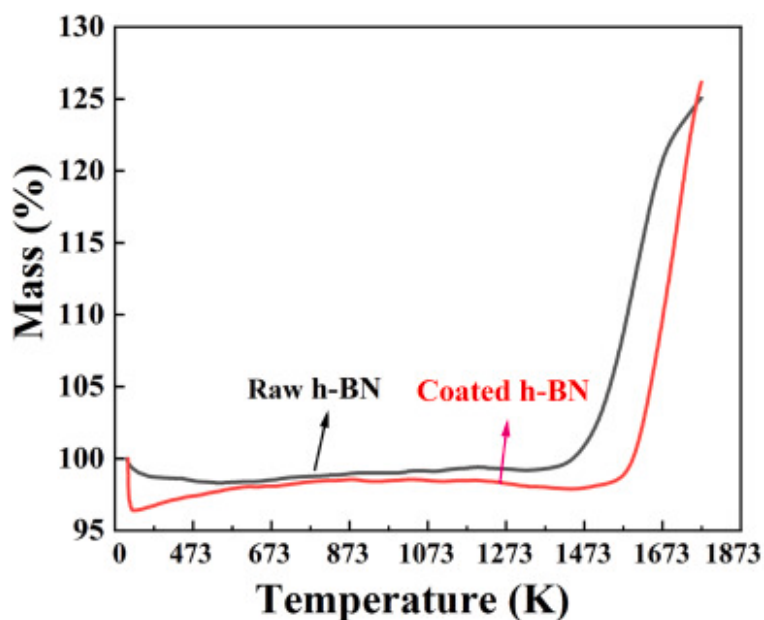
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Fig. 4. Water contact angles of (a) raw *h*-BN, and coated ones prepared (b) without surfactants, and with (c) SDBS and (d) PVP.

To visually test the dispersion behavior of coated *h*-BN, it was directly dispersed in water (Fig. S10). The raw *h*-BN floated on the water's surface, while the coated *h*-BN was uniform throughout the water (Fig. S10a–b). After 2 min stirring, the former one showed a layer of floating *h*-BN (Fig. S10c), and the boehmite-coated *h*-BN sank in the bottom of bottle, leaving a relatively clear liquid (Fig. S10d). This suggests that a boehmite-coating could not only confer the hydrophilicity on *h*-BN, promoting the wetting and dispersion in water, but also increase the density of *h*-BN, making it easily sink in water. Moreover, the sample surfaces exhibited no significant changes before and after ball milling (Fig. S11), both still being distributed with boehmite, and we also tested the water contact angle of the powders after ball milling (Fig. S12), which did not have any change, still showing the

hydrophilicity. From the perspective of application requirements, these results could demonstrate that the boehmite-coating should have good service strength in MgO castable.

To evaluate the oxidation resistance of as-prepared boehmite-coated *h*-BN, TG analyses were carried out in both original *h*-BN and coated one under air atmosphere. As shown in Fig. 5, the weight loss observed at 373K was attributed to the removal of adsorbed water. The onset oxidation temperature of pristine *h*-BN was approximately 1327K, whereas that of the coated one was elevated to 1573K. The boehmite coating could form a dense and stable "shell" on *h*-BN surface, which acts as a physical barrier that impedes oxygen, thus enhancing the oxidation resistance.



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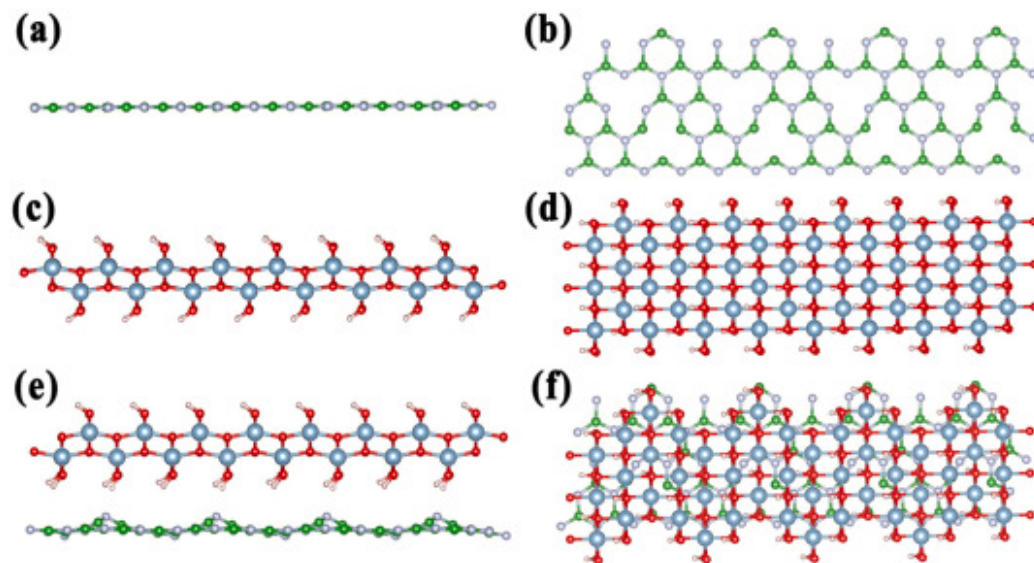
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Fig. 5. TG curves of raw *h*-BN and boehmite coated *h*-BN composite powders.

3.2. Theoretical study of water wettability of boehmite-coated *h*-BN

To clarify the wetting mechanism of coated *h*-BN, the adsorption of H₂O molecules onto the *h*-BN face are further considered based on DFT calculations. The adsorption structures of H₂O molecule on original *h*-BN, boehmite and boehmite-coated one were

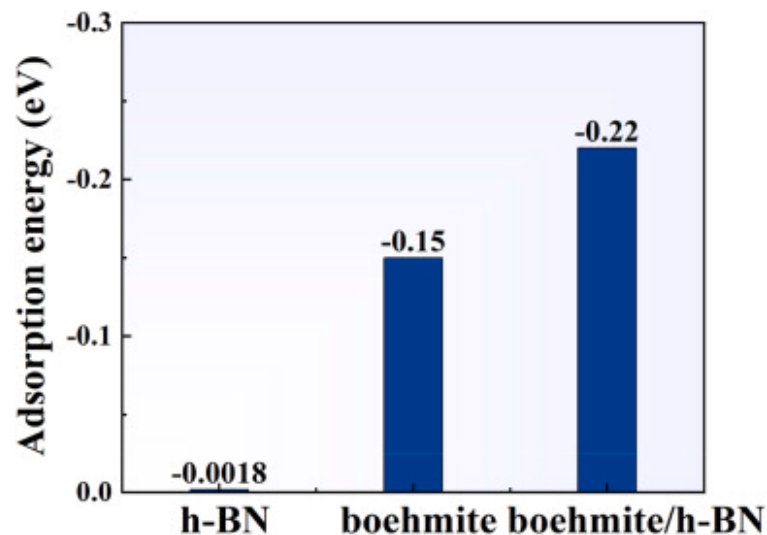
optimized, respectively (Fig. 6 and S13), and the adsorption energies were calculated. As shown in Fig. 7, the energy value on *h*-BN was -0.0018 eV , while it distinctly increased to -0.15 eV on boehmite. Furthermore, the H_2O adsorption energy on the boehmite-coated *h*-BN surface was increased to -0.22 eV , indicating that boehmite could enhance the adsorption capacity for H_2O molecules on *h*-BN surface [36].



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Fig. 6. Geometric structures of single H_2O adsorbed on (a, b) *h*-BN, (c, d) boehmite and (e, f) boehmite/*h*-BN.

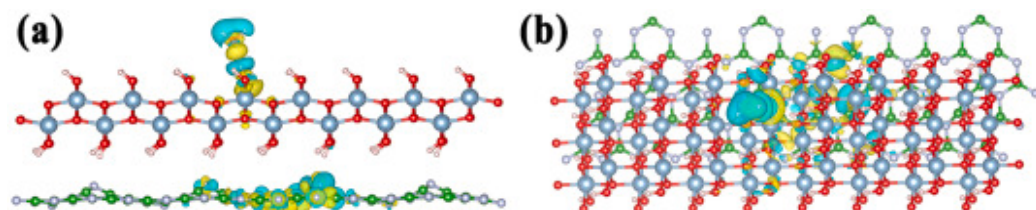


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Fig. 7. Adsorption energies of H₂O molecule on the surface of original *h*-BN, boehmite and boehmite-coated one.

Fig. 8 presents the charge density of H₂O molecules on boehmite/*h*-BN. It could be seen that a clear electronic localization phenomenon could be observed, where some charges on the O atoms of H₂O molecule were transferred to the H atoms of boehmite, resulting in a decrease of O negative charge in H₂O molecule. The bond lengths of H₂O molecule were stretched after adsorption, demonstrating that the adsorption onto boehmite/*h*-BN would weaken the two H-O bonds, i.e., the adsorbed H₂O molecule should be activated, which made them easier to be adsorbed and thus improved the wettability of boehmite-coated *h*-BN [37,38].



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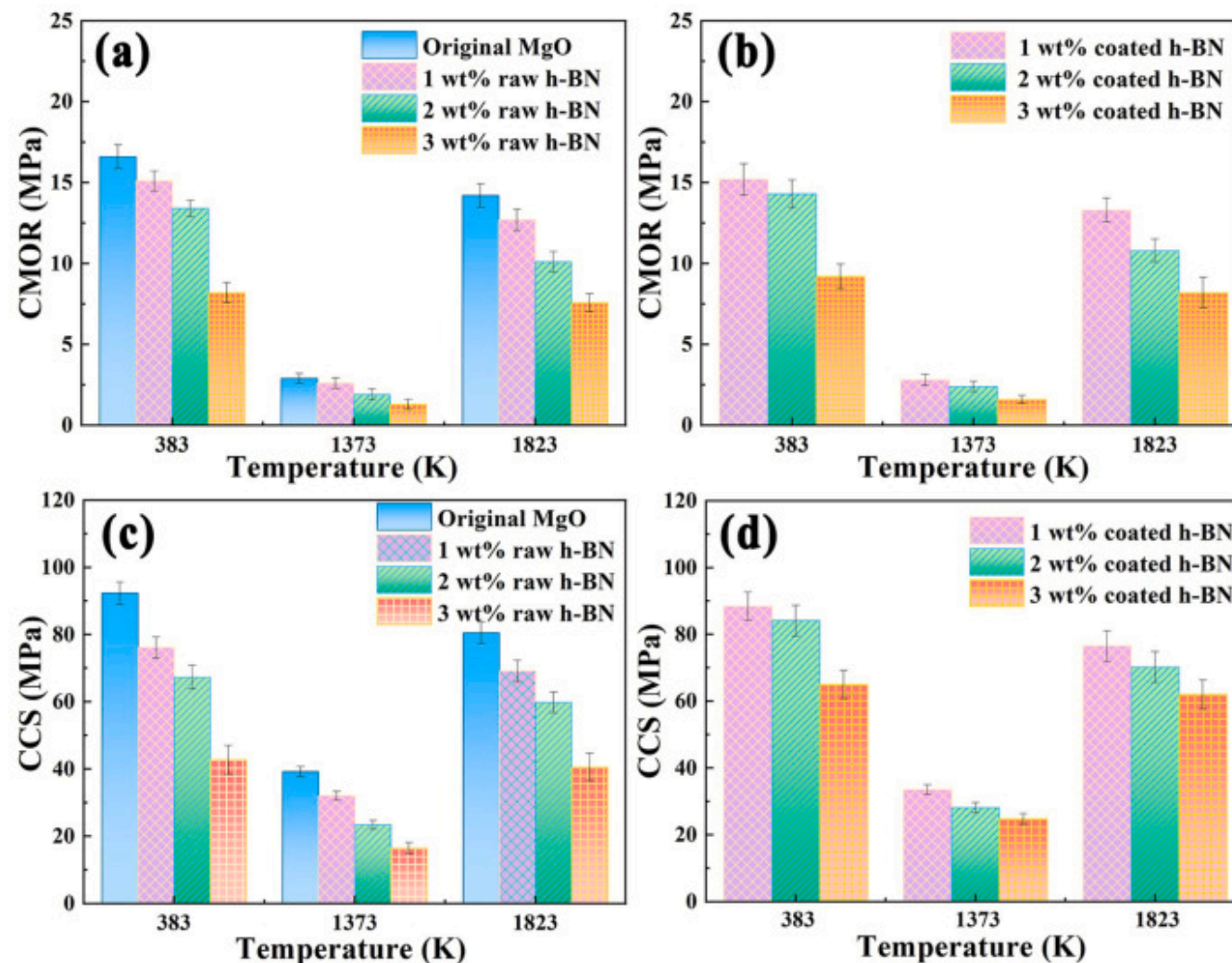
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Fig. 8. Charge density difference of H₂O absorbed boehmite/h-BN.

3.3. Preparation and mechanical properties of *h*-BN modified MgO castables

Fig. S14 presents the water additions for MgO castable containing different amounts of coated *h*-BN (a consistent flowability of 175 ± 2.5 mm). For the specimen without using *h*-BN, the water addition was only 5.2 wt%. When the specimen contained 1, 2 and 3 wt% uncoated *h*-BN, the water content was respectively 5.6, 6.4 and 7.3 wt%. In comparison, they could be reduced to 5.4, 5.7 and 6.7 wt%, respectively, in the case of using boehmite-coated *h*-BN. Fig. S15 shows the corresponding characterizations of castables following drying for 24 h at 383 K. The apparent porosity increased while bulk density decreased with the *h*-BN addition, while the sample using coated *h*-BN displayed a lower apparent porosity and higher bulk density (Fig. S15a and b). When firing the samples at 1373 K, it led to an obvious increase in the apparent porosity (Fig. S15c and d), probably due to the decomposition of hydration products. When further increasing the temperature to 1823 K, it demonstrates that the apparent porosity decreased and bulk density increased (Fig. S15e and f). This might be attributed to: 1) the volume expansion caused by the generation of Mg-Al spinel, which filled the pores of castable, and 2) partial sintering accounted for the densification of castable. Notably, the samples containing coated *h*-BN showed the lower apparent porosities and higher bulk densities than the counterparts using an identical amount of raw *h*-BN.

Fig. 9 presents the CMOR and CCS of castables after 3 h firing at various temperatures. For samples without *h*-BN, the CMOR value exceeded 16 MPa following drying at 383 K but declined to 2.9 MPa after heat treatment at 1373 K (Fig. 9a) because of the dehydrated alumina [39]. After firing at 1823 K, the CMOR rose dramatically to 16.1 MPa due to the partial sintering and formation of Mg-Al spinel phases. Notably, the sample using 3 wt% raw *h*-BN exhibited a lowest CMOR, due to the high water addition and weak interfacial bonding between *h*-BN and matrix. In comparison, the specimen with the same content of coated *h*-BN demonstrated improvements in CMOR of 12.2%, 23.1%, and 7.9% after firing at 383 K, 1373 K, and 1823 K, respectively (Fig. 9b). Similarly, the CCS of coated *h*-BN modified samples were improved by 51.9%, 50.3% and 52.7%, respectively (Fig. 9c and d), compared with the counterparts with raw *h*-BN.



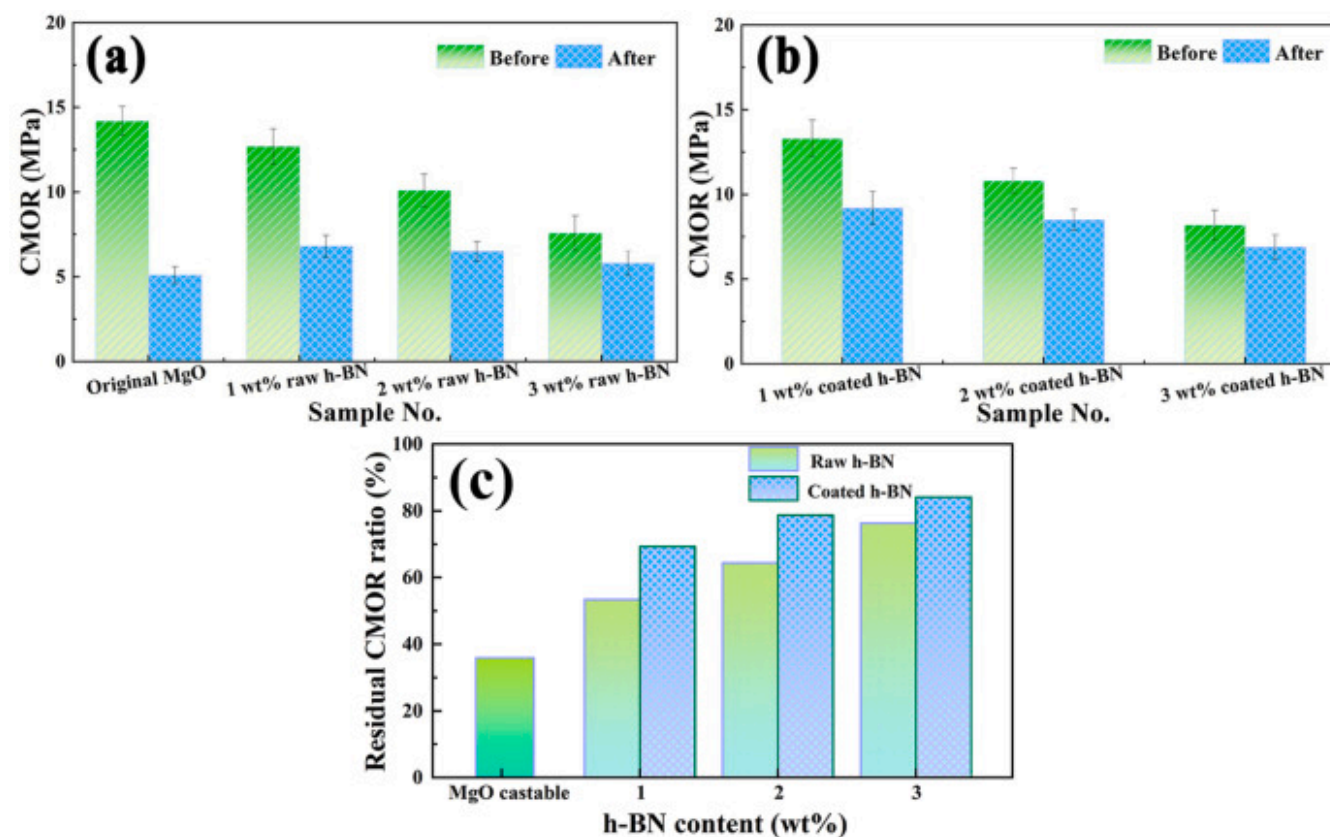
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Fig. 9. (a, b) Cold modulus of ruptures and (c, d) cold compressive strength of *h*-BN-modified MgO castables fired at different temperatures.

Thermal shock test was then conducted on *h*-BN-modified MgO castables. As shown in Fig. 10a and b, all samples experienced a notable decrease in CMOR following thermal shock. However, the residual CMOR ratio improved from 76.3% to 84.1% with the

addition of 3 wt% coated *h*-BN, indicating that boehmite coating on *h*-BN enhanced the thermal shock resistance of castables (Fig. 10c). This could be attributed to the less formation of microcracks, which resulted from an uniform dispersion of coated *h*-BN within the matrix, as verified by SEM (Fig. S16).



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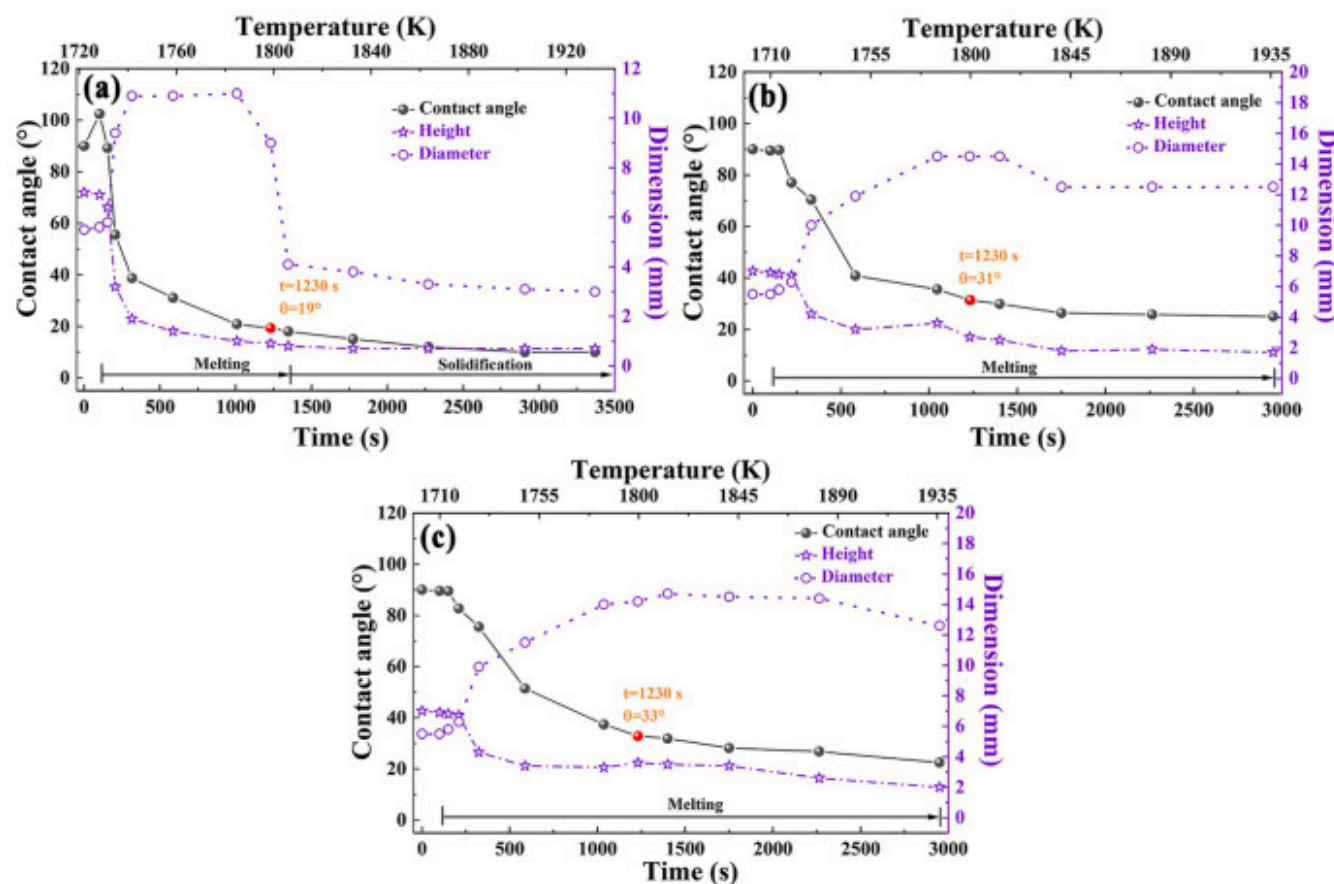
Fig. 10. CMOR change before and after (a, b) thermal shock and (c) residual ratio of samples with different *h*-BN contents.

Fig. S17 shows the cross-sectional images of castables after the static crucible tests, and Table S4 measured the corresponding corrosion index. It demonstrates that the slag resistance initially increased and then decreased with *h*-BN addition. When 1 wt% of raw *h*-BN was added, the corrosion index reached 40%, being significantly lower than that of sample without *h*-BN (55%, Fig. S17a). This improvement is attributed to the chemical inertness and slag non-wettability of *h*-BN, which could hinder the slag

corrosion and penetration [40]. When increasing the raw *h*-BN content to 2wt%, it further reduced the index to 29%. However, the corrosion index increased to 36% with 3 wt% of *h*-BN. A similar trend was observed in the case of using coated *h*-BN, and a lowest corrosion index of 23% was recorded in the sample containing 2wt% of coated *h*-BN.

3.4. Wetting behavior between slag and *h*-BN modified MgO castables

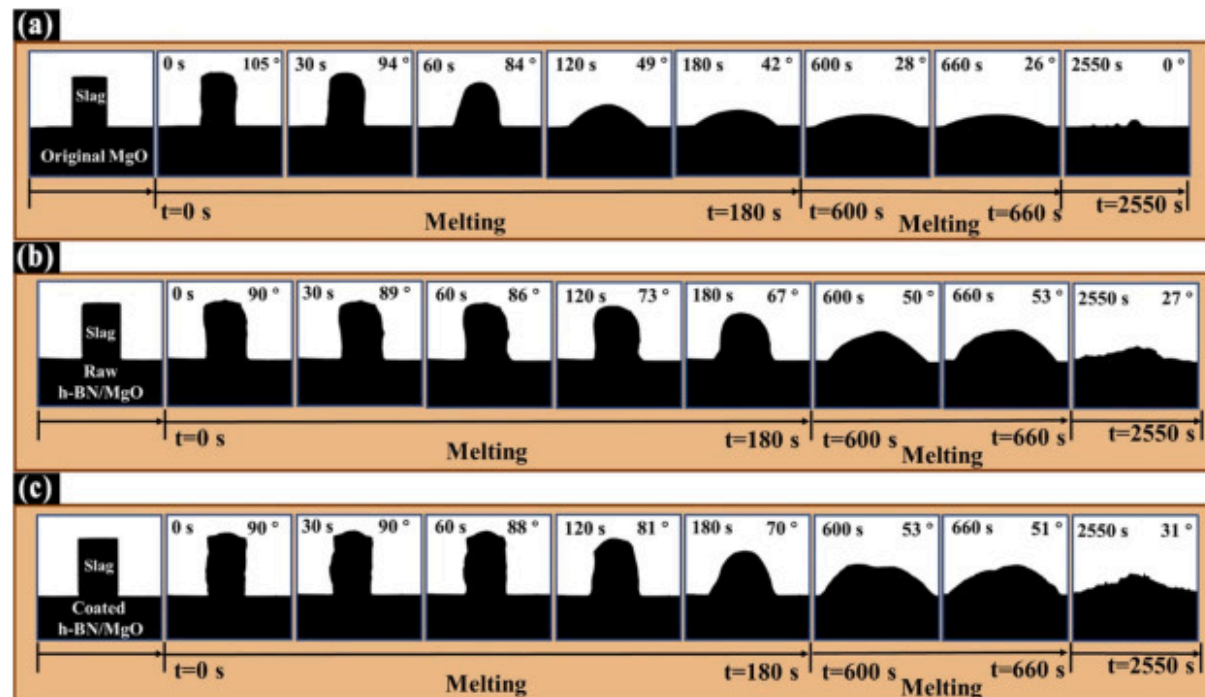
The slag wetting behaviors of MgO castables without and with 2wt% of raw *h*-BN and coated *h*-BN were investigated *via* sessile drop test. With increasing temperature, the solid slag ingot gradually changed shape from a cube to a spherical crown due to its melting (Fig. 11), allowing for qualitative evaluation of slag wettability based on the droplet morphology (Fig. 12).



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Fig. 11. Wetting curves of slag on the surface of MgO castables (a) without *h*-BN, and with (b) 2wt% raw *h*-BN and (c) 2wt% coated *h*-BN.



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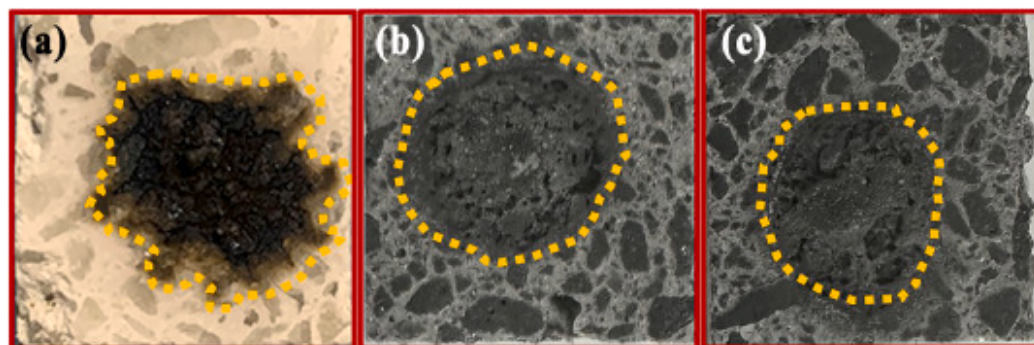
Fig. 12. Appearance i of slag on the surface of various samples: (a) original MgO sample, (b) 2wt% raw *h*-BN **modified MgO** castables and (c) 2wt% coated *h*-BN **modified MgO** castables.

The slag wetting curves parameters including contact angle (θ), height (h) and diameter (d) of slag ingots were recorded to investigate the wetting behavior. As shown in Fig. 11, they all underwent a sequential slag melting and solidification process. The slag transformation from the solid to liquid phase resulted in the appreciable changes in θ , d , and h . For the MgO castable without *h*-BN (a porosity of 17.5%, Fig. 11a), upon slag melting, θ and h notably decreased, while d expanded over time, reflecting a rapid wetting behavior. A steady contact angle of $\sim 19^\circ$ was observed at $t=1230$ s, reflecting the wettability of the sample toward the slag melt. In contrast, the slag spread more slowly on the MgO sample with 2wt% raw *h*-BN (Fig. 11b), the contact angle was $\sim 31^\circ$

at 1230s. For the sample with 2wt% coated h-BN, the slag spreading was further retarded, resulting in a contact angle of 33° at 1230s. This should be ascribed to a more homogeneous dispersion of boehmite-coated *h*-BN in the castable (Fig. 11c).

Fig. 12a demonstrates that the slag spreads rapidly on the sample without *h*-BN. The hemispherical slag droplet decreased rapidly over time, with the contact angle decreasing from 105° to 84° within the first 60s. It further decreased to 49° in the next 60s, representing a reduction of 41.7%, and then gradually decreased to 42° in the following 60s. At 2550s, the slag almost completely infiltrated the interior of the MgO sample, and the contact angle decreased to 0°. In contrast, the slag spread more slowly on the surface of the MgO sample containing 2wt% raw h-BN (Fig. 12b). The contact angle decreased from 89° to 86° within the first 60s and further decreased to 73° in the subsequent 60s, corresponding to a reduction of 15.1%. At 2550s, the contact angle of the slag on this sample remained at 27°. For the sample containing 2wt% coated *h*-BN composite powder (Fig. 12c), the softening of the slag into a hemispherical shape on its surface was further delayed (a reduction of 7.9%) with a contact angle of 53° at 600s. At 2550s, the contact angle between the slag and the sample was still 31°. These results indicate that the incorporation of the coated *h*-BN composite powder enhances the non-wetting behavior of the castables to the slag.

Slag infiltration can be directly studied *via* the top view of sample after high-temperature wetting experiment. From Fig. 13 and Table 1, the slag completely penetrated into the MgO castable without *h*-BN, showing a black infiltrated circle with an area of ~235.4mm² (Fig. 13a). In contrast, a large amount of residual slag accumulated on the surface of 2wt% raw *h*-BN/MgO sample, and a corrosion area of ~217.1mm² was observed (Fig. 13b). For the sample with 2wt% coated *h*-BN, the corrosion area was further reduced to 184.9mm² (Fig. 13c). Among these three samples, the last one exhibited a minimum corrosion index of 30%, which was 21.1% lower than that of MgO castable without *h*-BN (Table 1).



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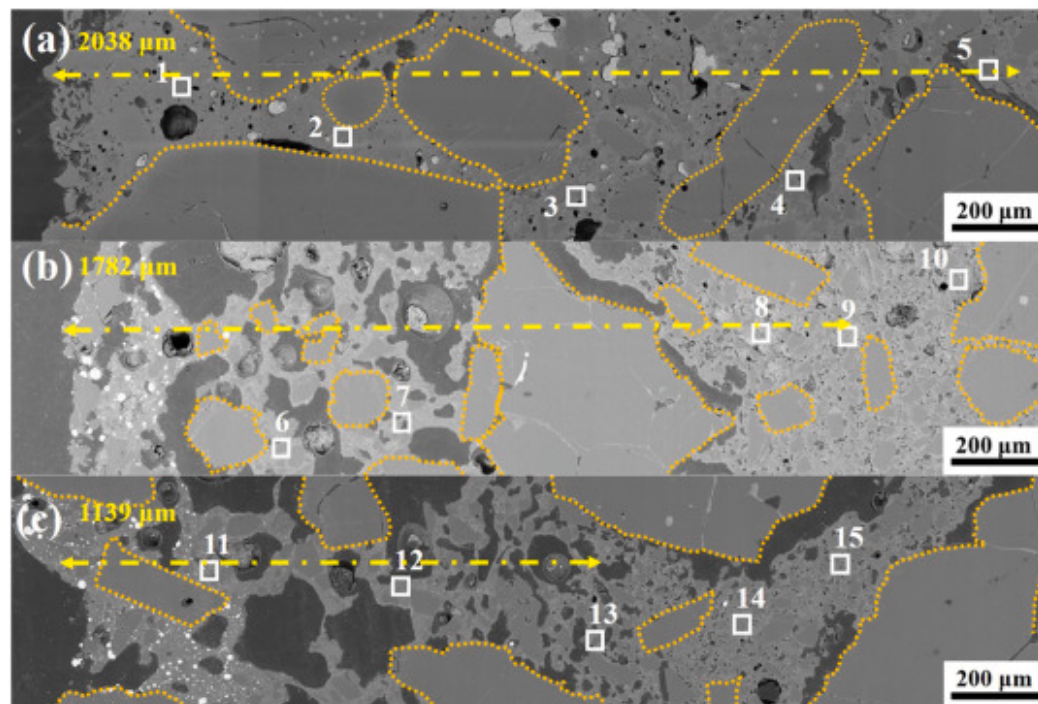
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Fig. 13. Top views of MgO castables (a) without *h*-BN, and with (b) 2 wt% raw *h*-BN and (c) 2 wt% coated *h*-BN after slag wetting experiment.

Table 1. Corrosion index of MgO castables (a) without *h*-BN, and with (b) 2 wt% raw *h*-BN and (c) 2 wt% coated *h*-BN.

Sample	Sample area (mm ²)	Area of corrosion (mm ²)	Corrosion index (%)
(a) Raw MgO	625	235.4	38
(b) 2 wt% raw <i>h</i> -BN/MgO	625	217.1	35
(c) 2 wt% raw <i>h</i> -BN/MgO	625	184.9	30

To visualize the slag corrosion on these samples, EDS mapping was performed. As shown in [Fig. S18](#), partial enrichment of Si, Ca, and Al within the matrix of sample without *h*-BN, along with the well-defined boundaries of large MgO particles, suggesting that the matrix served as the primary pathway for slag corrosion. The slag phase elements were also observed at the interfaces of samples with both 2 wt% of raw and coated *h*-BN. The corresponding corrosion depths were standardized and given in [Fig. 14](#) and [Table S5](#). It demonstrates that the depths of as-prepared raw *h*-BN/MgO and coated *h*-BN/MgO samples were 1782 μm and 1139 μm, respectively, which is much lower than the case of pure MgO castable (a value of 2038 μm), demonstrating superior slag corrosion resistance of *h*-BN modified castables.



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Fig. 14. BSE images and dot-EDS of MgO castables (a) without *h*-BN, and with (b) 2wt% raw *h*-BN and (c) 2wt% coated *h*-BN after slag wetting experiments.

4. Conclusions

In summary, boehmite-coated *h*-BN was successfully synthesized *via* hydrothermal treatment, and its WCA could be reduced to 46° from 150° (in the case of uncoated counterpart), demonstrating that boehmite-coating could markedly improve the water wettability of *h*-BN. **DFT calculations showed that the** adsorptions of H₂O onto boehmite/*h*-BN weaken the H-O bonds, which make water easier to be adsorbed, thus improving the wettability of boehmite-coated *h*-BN. **Owing to the hydrophilicity and slag non-wettability of coated *h*-BN**, the high-temperature slag contact angle of modified MgO castable was ~33°, which is higher than that of original one (~19°). As a result, the slag penetration depth of modified castable was reduced by 44.1%, and it

also exhibited a better residual CMOR ratio after thermal shock. This work not only achieves an even dispersion of water-repellent powders in castable system, but also provide a novel carbon-free refractory for metallurgical production.

CRedit authorship contribution statement

Yage Li: Writing – review & editing, Writing – original draft. **Jianli Wu:** Software. **Junyi Lv:** Validation. **Keke Guan:** Supervision. **Jie Song:** Conceptualization. **Zhong Huang:** Writing – review & editing. **Haijun Zhang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is/are the supplementary data to this article:

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Multimedia component 1.

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